COMPOSITION OF FOURIER-MUKAI FUNCTORS IS FOURIER-MUKAI

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- QCoh preservation. lemma-fourier-Mukai-QCoh (same chapter, judged citation-only, not carded). Makes the statement meaningful.
- Base change and projection formula. perfect-lemma-compare-base-change, perfect-lemma-cohomology-base-change (cross-chapter, named). Drive the equalities chain.

Source reference. Original lemma 0FYR: sources/stacks-equiv/pinned-original-source.tex, source lines 1171–1171. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08EU: focused-seeds/geometry/S9-eqv-compose-fourier-mukai/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08IB: focused-seeds/geometry/S9-eqv-compose-fourier-mukai/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0FXW: focused-seeds/geometry/S9-eqv-compose-fourier-mukai/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08D5: focused-seeds/geometry/S9-eqv-compose-fourier-mukai/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08DW: focused-seeds/geometry/S9-eqv-compose-fourier-mukai/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08DX: focused-seeds/geometry/S9-eqv-compose-fourier-mukai/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01K5: focused-seeds/geometry/S9-eqv-compose-fourier-mukai/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01KU: focused-seeds/geometry/S9-eqv-compose-fourier-mukai/ref-tags.txt, source lines 12–12. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X, Y, Z be schemes over S. Assume $X \to S$, $Y \to S$, and $Z \to S$ are quasi-compact and quasi-separated. Let $K \in D_{QCoh}(\mathcal{O}_{X \times_S Y})$. Let $K' \in D_{QCoh}(\mathcal{O}_{Y \times_S Z})$. Consider the Fourier-Mukai functors $\Phi_K : D_{QCoh}(\mathcal{O}_X) \to D_{QCoh}(\mathcal{O}_Y)$ and $\Phi_{K'} : D_{QCoh}(\mathcal{O}_Y) \to D_{QCoh}(\mathcal{O}_Z)$. If X and Z are tor independent over S and $Y \to S$ is flat, then*

$$\Phi_{K'} \circ \Phi_K = \Phi_{K''} : D_{QCoh}(\mathcal{O}_X) \longrightarrow D_{QCoh}(\mathcal{O}_Z)$$

where

$$K'' = Rpr_{13,*}(Lpr_{12}^* K \otimes^{\mathbf{L}}_{\mathcal{O}_{X \times_S Y \times_S Z}} Lpr_{23}^* K')$$

in $D_{QCoh}(\mathcal{O}_{X \times_S Z})$.

Proof. The statement makes sense by Lemma 0FYR. We are going to use Derived Categories of Schemes, Lemmas 08DW, 08DX, and 08D5 and Schemes, Lemmas 01K5 and 01KU without further mention. By Derived Categories of Schemes, Lemma 0FXW we see that $X \times_S Y$ and $Y \times_S Z$ are tor independent over Y. This means that we have base change for the cartesian diagram

$$\begin{array}{ccc} X \times_S Y \times_S Z & \longrightarrow & Y \times_S Z \\ \downarrow & & \downarrow p_Y^{YZ} \\ X \times_S Y & \xrightarrow{p_Y^{XY}} & Y \end{array}$$

for complexes with quasi-coherent cohomology sheaves, see Derived Categories of Schemes, Lemma 08IB. Abbreviating $p^* = Lp^*$, $p_* = Rp_*$ and $\otimes = \otimes^{\mathbf{L}}$ we have for

$M \in D_{QCoh}(\mathcal{O}_X)$ the sequence of equalities

$$\begin{aligned}
\Phi_{K'}(\Phi_K(M)) &= p^{YZ}_{Z,*}(p^{YZ,*}_Y p^{XY}_{Y,*}(p^{XY,*}_X M \otimes K) \otimes K') \\
&= p^{YZ}_{Z,*}(\mathrm{pr}_{23,*}\mathrm{pr}^*_{12}(p^{XY,*}_X M \otimes K) \otimes K') \\
&= p^{YZ}_{Z,*}(\mathrm{pr}_{23,*}(\mathrm{pr}^*_1 M \otimes \mathrm{pr}^*_{12} K) \otimes K') \\
&= p^{YZ}_{Z,*}(\mathrm{pr}_{23,*}(\mathrm{pr}^*_1 M \otimes \mathrm{pr}^*_{12} K \otimes \mathrm{pr}^*_{23} K')) \\
&= \mathrm{pr}_{3,*}(\mathrm{pr}^*_1 M \otimes \mathrm{pr}^*_{12} K \otimes \mathrm{pr}^*_{23} K') \\
&= p^{XZ}_{Z,*}\mathrm{pr}_{13,*}(\mathrm{pr}^*_1 M \otimes \mathrm{pr}^*_{12} K \otimes \mathrm{pr}^*_{23} K') \\
&= p^{XZ}_{Z,*}(p^{XZ,*}_X M \otimes \mathrm{pr}_{13,*}(\mathrm{pr}^*_{12} K \otimes \mathrm{pr}^*_{23} K'))
\end{aligned}$$

as desired. Here we have used the remark on base change in the second equality and we have use Derived Categories of Schemes, Lemma 08EU in the 4th and last equality. □